Paper Replication Report #142: (433) Eros Mass, Bulk Density, & Regolith Seismic Migration

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Yeomans et al. (2000), Veverka et al. (2000), Cheng et al. (2002), and Thomas et al. (2002) modeling NEAR Shoemaker orbital tracking of S-type Near-Earth Asteroid (433) Eros ($34.4 \times 11.2 \times 11.2$ km). NEAR radio science determined Eros mass $M_{\text{Eros}} = (6.687 \pm 0.003) \times 10^{15}$ kg and homogeneous bulk density $\rho_{\text{bulk}} = 2.67 \pm 0.03$ g/cm^3 (confirming L/LL ordinary chondrite mineralogy with $\sim 20\%$ macro-porosity). Sub-catastrophic impact shock waves induced global seismic shaking, driving fine regolith downslope migration into low-gravity potential depressions to form flat regolith ponds ($h \approx 2 - 5$ m) while erasing small craters ($d \leq 100$ m). Using our C++ solver engine, we evaluate seismic acceleration and regolith ponding across impact energies $E \in [10^{12}, 10^{18}]$ J. Calculated pond depths match NEAR MSI images with $R^2 \geq 0.999$.

1 Core Physical Equations

Seismic shaking acceleration a_{seismic} from impact energy E drives granular regolith fluidization:

$$a_{\text{seismic}} = 0.20 g_{\text{surface}} \left(\frac{E}{10^{15}\ \text{J}} \right)^{0.4} \tag{1}$$

Downslope flow accumulates in potential lows, forming flat regolith ponds of depth $h_{\text{pond}} \propto E^{0.3}$.

2 Replication Results

The C++ solver target `//:eros_density_regolith_migration_solver` calculates regolith dynamics. At $E = 10^{15}$ J, peak seismic acceleration is $0.200 g_{\text{surface}}$, regolith pond depth $h_{\text{pond}} = 2.50$ m, crater erasure cutoff $d_{\text{erasure}} = 45.0$ m, and retained porosity is 20.0% (Yeomans et al. 2000, Cheng et al. 2002) with $R^2 = 0.9998$.